

UWBParserThread Documentation

DEVELOPER REFERENCE MANUAL — CORE API SUBSET

This document provides targeted technical documentation for the fundamental components of the `UWBParserThread` class. This subset focuses exclusively on the object lifetime initialization mechanism and the thread-safe API used to query spatial coordinates.

1. Class Initialization: `__init__()`

The constructor initializes the background execution context, establishes thread locking primitives, and configures either explicit or discovery-based serial communication channels linked to the Ultra-Wideband (UWB) transceiver hardware.

```
def __init__(self, x_origin=0.0, y_origin=0.0, serial_port=None, baud_rate=921600):
    super().__init__()
    self.serial_port = serial_port if serial_port else self.detect_com_port()
    self.baud_rate = baud_rate
    self.data_lock = threading.Lock()
    self.tag_data = {} # Stores tag data {tag_id: (x, y, update_time)}
    self.origin_x = 0.0
    self.origin_y = 0.0
    self.running = True
```

Input Parameters

Parameter	Type	Default	Description
<code>x_origin</code>	<code>float</code>	<code>0.0</code>	Defines the baseline coordinate offset along the spatial X-axis.
<code>y_origin</code>	<code>float</code>	<code>0.0</code>	Defines the baseline coordinate offset along the spatial Y-axis.
<code>serial_port</code>	<code>str</code> <code>None</code>	<code>None</code>	Explicit device port name (e.g., 'COM3'). If None, the automated hardware auto-discovery routine is executed.
<code>baud_rate</code>	<code>int</code>	<code>921600</code>	The transmission clock frequency rate for physical UART serial framing.

2. Thread-Safe Query Interface: `get_tag_position()`

This method provides an asynchronous, thread-safe gateway for consumer applications to pull the most recent localization calculations from the streaming hardware dictionary instance matrix without creating race conditions.

```
def get_tag_position(self, tag_id):  
    # Returns the x, y position and update time of a specific tag.  
    with self.data_lock:  
        return self.tag_data.get(tag_id, (None, None, None))
```

Input Parameters

Parameter	Type	Description
<code>tag_id</code>	<code>int</code>	The unique physical identifier sequence assigned to the tracking target peripheral board.

Output / Return Values

The method returns an atomic, un-ordered 3-element collection `tuple` containing scaled geometric coordinates in metric dimensions mapped against the processing system timestamp framework:

Return Format: `(pos_x, pos_y, update_time)`

- `pos_x` : `float` | `None` — Deserialized position value along the X-axis translated directly to meters. Returns `None` if the identifier is invalid or missing.
- `pos_y` : `float` | `None` — Deserialized position value along the Y-axis translated directly to meters. Returns `None` if the identifier is invalid or missing.
- `update_time` : `float` | `None` — Absolute Unix timestamp representation matching the exact execution loop frame window where the packet payload validation concluded.